EXAM SOLUTIONS

1. **These are mostly basic questions directly form slides and course materials, or problems similar to those solved in the class, and require an understanding of basic concepts.**

[3]

In non-coherent demodulation of AM-modulated signals, when the carrier-to-noise ratio falls below unity, the noise term starts dominating the performance, and the performance of the envelop detector deteriorates very rapidly. This is not the case in coherent demodulation.

ii) [3]

It is impossible to demodulate PSK with a non-coherent detector, e.g., envelop detection, since PSK signals have the same frequency and amplitude. However, we can demodulate PSK signals differentially, where phase reference is provided by a delayed version of the signal in the previous interval. This is called differential phase shift keying (DPSK).

iii) [2]

Since it is white, its samples are uncorrelated:

$$E[X(t_1)X(t_2)] = R_N(t_2 - t_1) = 0, \text{ for } t_1 \neq t_2.$$

Since it is also Gaussian, samples are independent.

iv) [2]

Entropy is only a function of the probability mass function, not the particular values the random variable takes. Therefore, $H(Y) = H(X) =$ 0.85 bits.

a) True. A stationary process has time-invariant distribution, and hence, a time-invariant mean. [2]

ii) True. WSS is preserved going through a linear time invariant (LTI) system. [2]

iii) False. Being Gaussian is about the distribution, it does not specify the power spectral density. [2]

iv) True. We have $E[X(t)Y(t)] = E[X(t)]E[Y(t)] =$ constant, and

$$\begin{aligned} R_{XY}(t_1, t_2) &= E[X(t_1)Y(t_1)X(t_2)Y(t_2)] \\ &= E[X(t_1)X(t_2)]E[Y(t_1)Y(t_2)] \\ &= R_X(t_2 - t_1)R_Y(t_2 - t_1). \end{aligned}$$

[3]

b) i) [2]

With $m_p = 3$, the maximum quantization error is given by $\Delta/2 = 2m_p/2L = 3/256 = 0.0117$.

ii) [3]

The SNR at the quantizer output is given by $SNR = \frac{3P}{m_p^2} 2^{2n}$, where P is the average power of the signal and n is the number of bits per sample, i.e., $n = \log_2 L$. We have $P = A_m^2/2 = 9/2$, where $A_m = 3$, and $n = 8$ bits. Therefore, we obtain $SNR = 3 \cdot 2^{15} \sim 49.9$ dB.

iii) [2]

Nyquist frequency is 8 kHz. The sampling frequency is 10 kHz, leading to the PCM bandwidth of $8 * 10 = 80$ kbits per second.

iv) [3]

If the sampling frequency is 14 kHz, we should have $14n = 80$; hence, we can have at most 5 bits per sample.

This would result in $SNR = 3 \cdot 2^9 \sim 31.9$ dB. Note that a reduction of 3 bits per samle results in 3x6=18 dB reduction in the SNR.

c) We have

$$\begin{aligned} S_X(f) &= \frac{\pi/2}{\left(\frac{\pi}{2}\right)^2 + \left(2\pi\left(f - \frac{1}{4}\right)\right)^2} + \frac{\pi/2}{\left(\frac{\pi}{2}\right)^2 + \left(2\pi\left(f + \frac{1}{4}\right)\right)^2} \\ &= \left[\frac{2(\pi/2)}{\left(\frac{\pi}{2}\right)^2 + (2\pi f)^2}\right] * \frac{1}{2}\left[\delta(f - \frac{1}{2}) + \delta(f + \frac{1}{4})\right] \end{aligned}$$

The autocorrelation of the given WSS random process can be found by taking the inverse Fourier transform of the given PSD.

$$R_X(\tau) = e^{-\frac{\pi}{2}|\tau|} \cdot \cos\left(\frac{\pi}{2}\tau\right).$$

For a real WSS process, we have

$$\begin{aligned} E\left[\left(X(t+\tau) - X(t)\right)^2\right] &= E[X^2(t+\tau) - 2X(t+\tau)X(t) + X^2(t)] \\ &= 2[R_X(0) - R_X(\tau)]. \end{aligned}$$

i) [5]

For $\tau = 1/\pi$, we have

$$2[R_X(0) - R_X(\tau)] = 2[1 - e^{-1/2}\cos(1/2)].$$

ii) [6]

$$\begin{aligned}
&\mathrm{E}\left[(X(t+1)+X(t+3)-X(t+5)-X(t))^2\right] \\
&\quad = \mathrm{E}[X^2(t+1)+X^2(t+3)+X^2(t+5)+X^2(t) \\
&\quad +2X(t+1)X(t+3)-2X(t+1)X(t+5)-2X(t+1)X(t) \\
&\quad -2X(t+3)X(t+5)-2X(t+3)X(t)+2X(t+5)X(t) \\
&= 4R_X(0)-2R_X(4)-2R_X(1)-2R_X(3)-2R_X(5).
\end{aligned}$$

We note that $R_X(1) = R_X(3) = R_X(5) = 0$ since $\cos(\tau\pi/2) = 0$ when τ is an odd integer. Therefore, the second moment is given by

$$4R_X(0)+2R_X(4) = 4-2e^{-2\pi}.$$

2. **The first part is similar to problems solved in the class, and expects a good understanding of the fundamental conceipts relating to random processes. The second one is a different problem that will require some deeper thinking and manipulation of the concepts learned in the lectures.**

a) [5]

We have

$$\begin{aligned}
E[X(t)] &= E[N_1\cos(2\pi f_c t) - N_2\sin(2\pi f_c t)] \\
&= E[N_1]\cos(2\pi f_c t) - E[N_2]\sin(2\pi f_c t) \\
&= 0
\end{aligned}$$

$$\begin{aligned}
R_X(t_1,t_2) &= E[X(t_1)X(t_2)] \\
&= E\left[(N_1\cos(2\pi f_c t_1) - N_2\sin(2\pi f_c t_1))(N_1\cos(2\pi f_c t_2) - N_2\sin(2\pi f_c t_2))\right] \\
&= E[N_1^2]\cos(2\pi f_c t_1)\cos(2\pi f_c t_2) + E[N_2^2]\sin(2\pi f_c t_1)\sin(2\pi f_c t_2) \\
&\quad - E[N_1N_2](\cos(2\pi f_c t_1)\sin(2\pi f_c t_2) + \sin(2\pi f_c t_1)\cos(2\pi f_c t_2)) \\
&= \sigma^2[\cos(2\pi f_c t_1)\cos(2\pi f_c t_2) + \sin(2\pi f_c t_1)\sin(2\pi f_c t_2)] \\
&= \sigma^2\cos(2\pi f_c(t_1-t_2)),
\end{aligned}$$

where we have used the linearity of the expectation operation, and the independence of N_1 and N_2, i.e., $E[N_1N_2] = E[N_1]E[N_2] = 0$. We can write the autocorrelation function as follows:

$$R_X(\tau) = \sigma^2\cos(2\pi f_c\tau).$$

ii) [2]

Since the mean is independent of time, and the autocorrelation function depends only on the time difference, $X(t)$ is WSS.

iii) [4]

We have

$$\begin{aligned} S_X(f) &= \mathbb{F}(R_X(\tau)) \\ &= \mathbb{F}\left(\sigma^2 \cos(2\pi f_c \tau)\right) \\ &= \frac{\sigma^2}{2}[\delta(f-f_c)+\delta(f+f_c)]. \end{aligned}$$

iv) [4]

We can consider $Y(t)$ as the output of an LTI filter with frequency response $H(f) = -j \cdot \text{sgn}(f)$. Then the PSD of $Y(t)$ is found as:

$$\begin{aligned} S_Y(f) &= |H(f)|^2 S_X(f) \\ &= |-j \cdot \text{sgn}(f)|^2 S_X(f) \\ &= \frac{\sigma^2}{2}[\delta(f-f_c)+\delta(f+f_c)]. \end{aligned}$$

b) i) [3]

In FSK, we have $y_1(t) = A + [n_1(t) - n_0(t)]$ and $y_0(t) = -A + [n_1(t) - n_0(t)]$. In this case the noise variance is given by $\sigma^2 = N_0/T_b + 2N_0/T_b = 3N_0/T_b$. Error probability is $Q\left(\frac{A}{\sigma}\right) = Q\left(\sqrt{\frac{2E_b}{3N_0}}\right)$.

ii) [4]

Error probability is $Q\left(\frac{3A}{2\sigma}\right)$.

Average energy per bit is $E_b = \frac{1}{2}\frac{A^2T_b}{2} + \frac{1}{2}\frac{4A^2T_b}{2} = \frac{5A^2T_b}{4}$.

The noise variance as before $\sigma^2 = 3N_0/T_b$. Therefore, the error probability can be obtained as $Q\left(\sqrt{\frac{3E_b}{5N_0}}\right)$.

This error probability is higher than the one in part i), because symmetric signalling is always more efficient in terms of average energy.

iii) [4]

We should transmit pulses of magnitudes A and $-A$ over frequency f_0, where the noise variance is lower. This would result in standard PSK with an error probability of $Q(A/\sigma)$, where $E_b = \frac{A^2T_b}{2}$ and $\sigma^2 = N_0/T_b$. Equivalently, the error probability is given by $Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$. This error probability is lower than both of the schemes in parts (i) and (ii).

iv) [4]

There are many ways to do this. We can transmit a pulse over frequency f_0 and another pulse over frequency f_1. Or, we can also use FSK as in part (i), but can have pulses of 4 different amplitudes, e.g., $(-3A, -A, +A, +3A)$.

3. **The first part is an unseen problem and requires the students to understand the impact of basic components of a demodulator, and connect it to the concepts learned in the class. Part b) has both a conceptual part and a more strightforwrd evaluation part. The last part can be answered based on basic probbaility knowledge, or directly using the answer to the conceptual question.**

a) [3]

We have

$$\begin{aligned} x_1(t) &= A\cos(2\pi ft)\cos(2\pi f_1 t) \\ &= \frac{A}{2}\cos(2\pi(f-f_1)t) + \frac{A}{2}\cos(2\pi(f+f_1)t) \end{aligned}$$

and

$$\begin{aligned} x_2(t) &= A\cos(2\pi ft)\sin(2\pi f_1 t) \\ &= -\frac{A}{2}\sin(2\pi(f-f_1)t) + \frac{A}{2}\sin(2\pi(f+f_1)t) \end{aligned}$$

ii) [4]

We have

$$y_1(t) = \frac{A}{2}\cos(2\pi(f-f_1)t)$$

and

$$y_2(t) = -\frac{A}{2}\sin(2\pi(f-f_1)t)$$

since $f+f_1 = f+(f_L+f_H)/2$ is greater than $W=(f_H+2f_L)/2$ for any $f \in [f_L, f_H]$.

iii) [3]

$$\begin{aligned} u(t) &= \frac{A}{2}\cos(2\pi(f-f_1)t)\cos(2\pi f_2 t) - \frac{A}{2}\sin(2\pi(f-f_1)t)\sin(2\pi f_2 t) \\ &= \frac{A}{2}\cos(2\pi(f_2+f-f_1)t) \end{aligned}$$

iv) [5]

This is upper single sideband modulation (SSB) with a carrier frequency of $f_2 - f_1$.

b) i) [4]

The joint entropy of two random variables is always smaller than or equal to the sum of their individual entropies. $H(X,Y)$ and $H(X)+H(Y)$ are equal if and only if X and Y are independent.

ii) [4]

The marginal distribution of X is given by $P(X=1)=P(X=2)=5/16$ and $P(X=3)=6/16$. Therefore, we have

$$\begin{aligned} H(X) &= \frac{5}{16}\log\frac{16}{5}+\frac{6}{16}\log\frac{16}{6} \\ &= 4-\frac{5}{8}\log 5-\frac{3}{8}(1+\log 3) \\ &= 1.579 \end{aligned}$$

Y has the exact same marginal distribution, and hence, the same entropy.

iii) [4]

The joint entropy can be calculated from the joint distribution as follows:

$$\begin{aligned} H(X,Y) &= 3\times\frac{1}{16}\log\frac{16}{1}+\frac{2}{16}\log\frac{16}{2}+\frac{3}{16}\log\frac{16}{3}+2\times\frac{4}{16}\log\frac{16}{4} \\ &= 4-\frac{1}{8}\log 2-\frac{3}{16}\log 3-\frac{8}{16}\log 4 \\ &= 2.578 \end{aligned}$$

iv) [3]

If they were independent, the joint entropy would be equal to the sum of individual entropies, which is not the case. Moreover, the fact that they are not independent is also evident from their joint distribution, which is not equal to the multiplication of their marginals.